

Mammals and birds

THESE GROUPS ARE WARM-BLOODED VERTEBRATES.

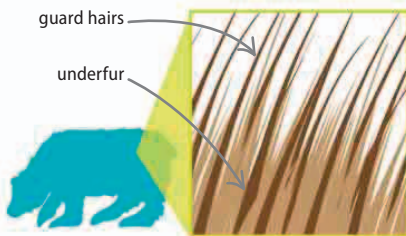
The vertebrate classes **Aves** (birds) and **Mammalia** (mammals) are among the most widespread groups of animal. They live on all continents and in almost all aquatic habitats.

Endothermy

Birds and mammals are endothermic (warm-blooded) animals, meaning they maintain a constant body temperature. This requires energy to warm or cool the body, but it ensures that the animal's metabolism runs at a constant rate. As a result, its body systems function fully—even in colder habitats where ectothermic (cold-blooded) animals cannot survive. Endotherms have anatomical features to help them manage their body heat.

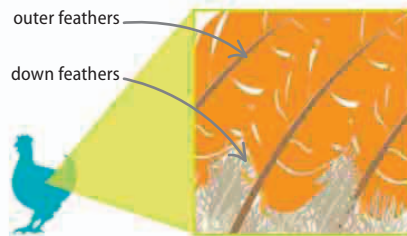
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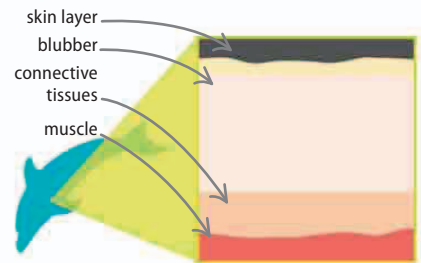
△ Fur layers

The hairs of many mammals are in two layers. The short underfur traps an insulating blanket of air. The longer, oily guard hairs keep out water, which would reduce the effectiveness of the underfur.



△ Down insulation

Birds prevent heat loss using fluffy down feathers that grow close to the body, under their outer feathers. Down traps air in pockets, insulating the body, and preventing valuable body heat from escaping.

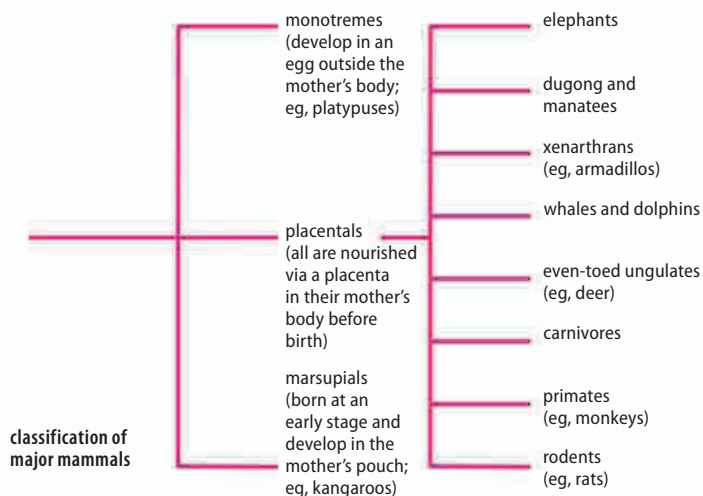


△ Blubber

In water, wet fur is a hindrance, so marine mammals have a thick insulating layer of blubber. This is a layer of soft fat, which has blood vessels running through it to help keep the animal warm.

Mammals

The largest vertebrates around today are mammals. The group gets its name from their mammary glands—modified sweat glands that produce milk. They are used by female mammals to suckle their young after birth. All mammals have at least a few hairs on their skin—although they are lost soon after birth in whales and dolphins. The hairs are made from keratin, the same waxy protein that builds reptile scales and bird feathers.



◁ **Mammal variety**
Mammals appeared around 200 million years ago (mya) as small insect-eaters similar to today's shrews. The great variety of species we see today evolved from these primitive ancestors after the dinosaurs became extinct 65 mya. By 30 mya, mammals were the dominant vertebrate group.